

Task 28: Implement Queue using Stacks

Problem Description

Implement a first in first out (FIFO) queue using only two stacks. The implemented queue should support all the functions of a normal queue (push, peek, pop, and empty).

Java Solution

```
class MyQueue {
    Stack<Integer> input;
    Stack<Integer> output;

    public MyQueue() {
        input = new Stack<>();
        output = new Stack<>();
    }

    public void push(int x) {
        input.push(x);
    }

    public int pop() {
        peek();
        return output.pop();
    }

    public int peek() {
        if (output.isEmpty()) {
            while (!input.isEmpty()) {
                output.push(input.pop());
            }
        }
        return output.peek();
    }

    public boolean empty() {
        return input.isEmpty() && output.isEmpty();
    }
}
```